INFORMATION TECHNOLOGY INSTITUTE

Analog IC Design Track

Analog Integrated Circuit Design

Lab 08

Negative Feedback

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PART 1: Feedback with Behavioral OTA

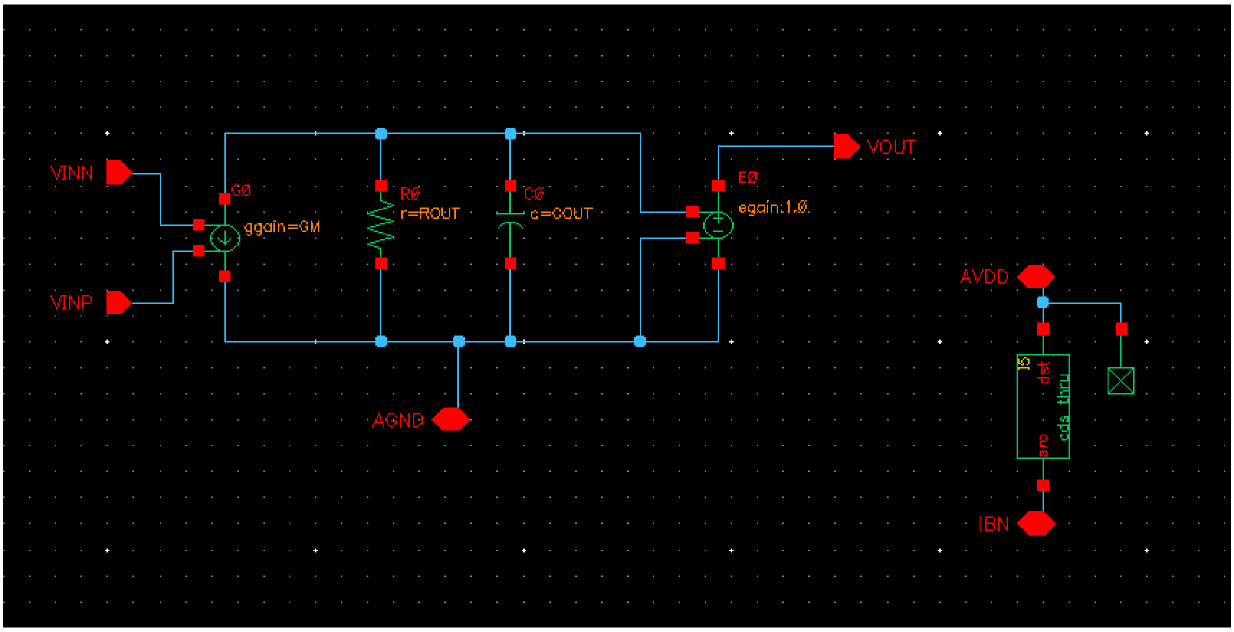

Figure 1: Behavioral Model Schematic

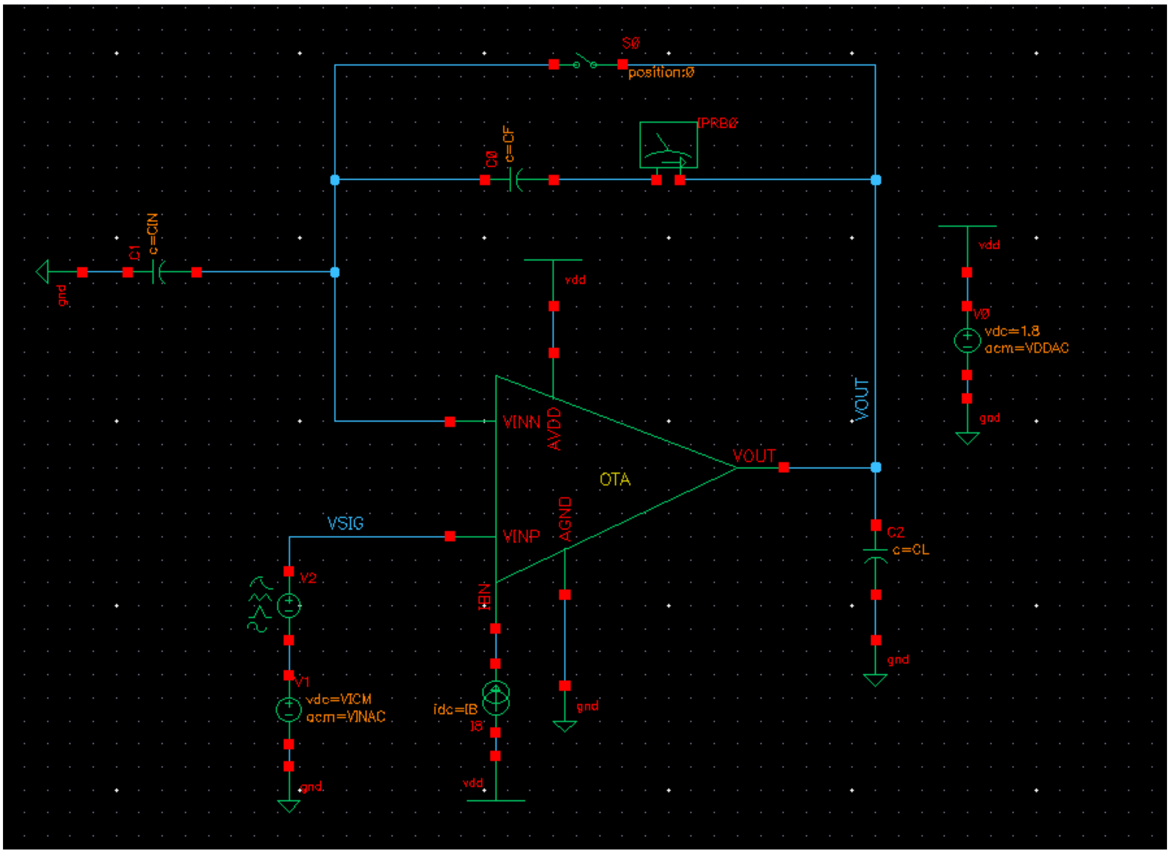

Figure 2: Testbench Schematic

$$R_{OUT} = \frac{A_v}{GM} = \frac{65}{157.3\,\mu A/V} = 414\ \mathrm{k\Omega}$$

$$C_{OUT} = \frac{GM}{\omega_u} = \frac{157.3\,\mu A/V}{2\pi(5.0081\,\mathrm{MHz})} = 5.0\ \mathrm{pF}$$

1.1 Closed-Loop Gain vs. Frequency

1.1.1 Simulation Results:

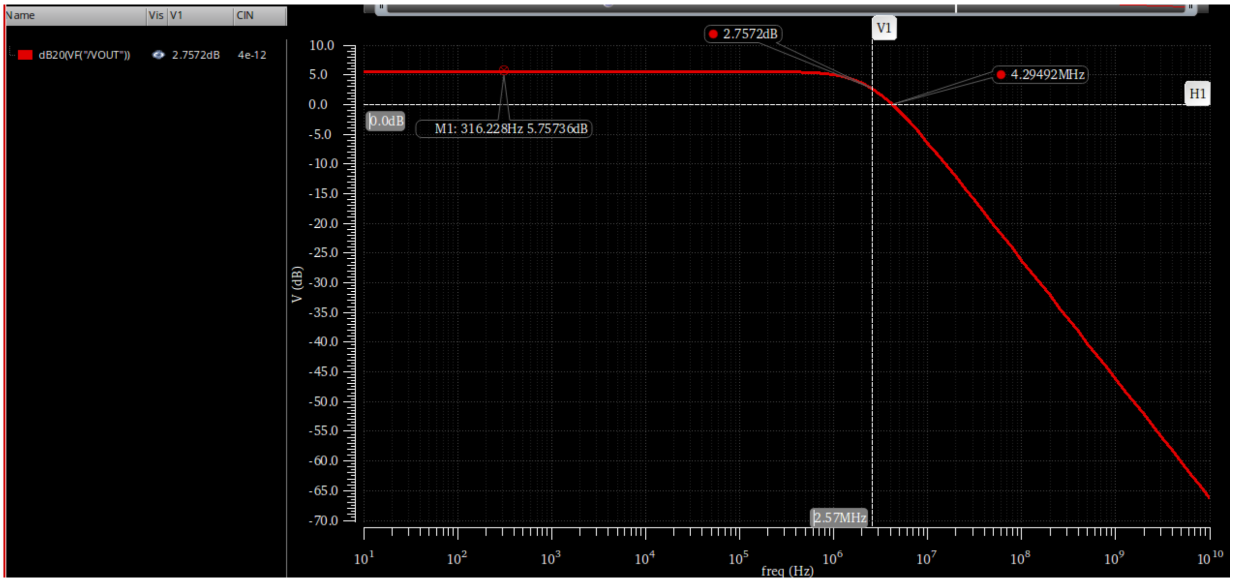

Figure 3: VOUT in dB with Cin=4pF

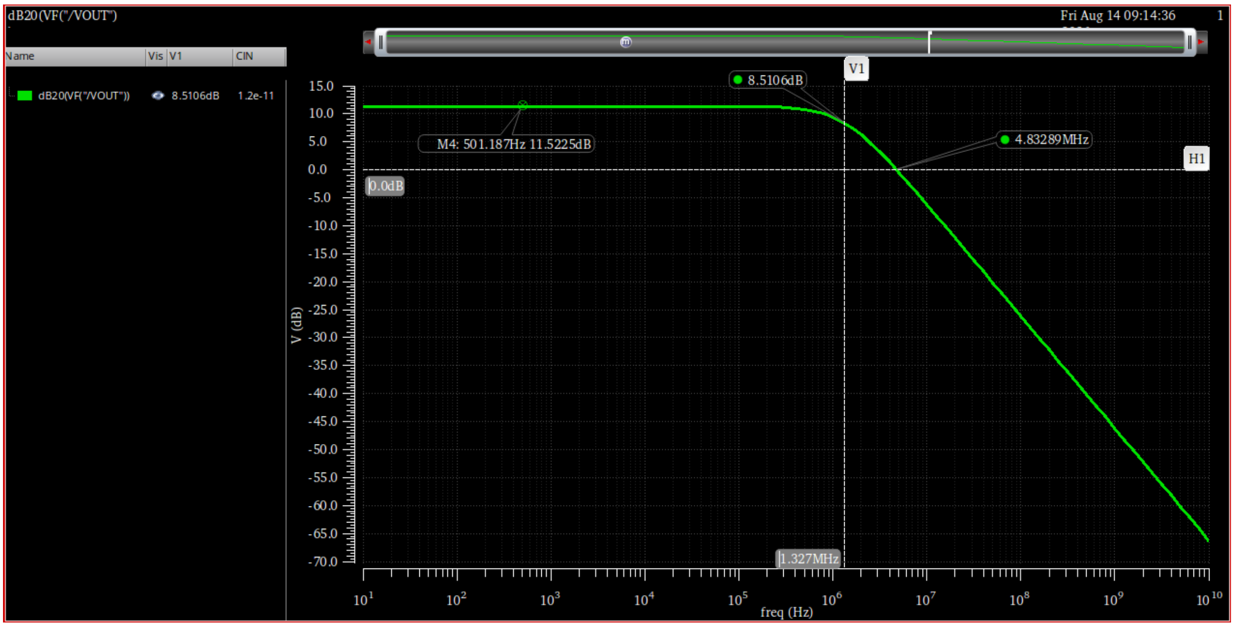

Figure 4: VOUT in dB with Cin=12pF

Parameters: CIN=4p			
1	ITI_LABS:LAB_08_tb:1	dB20(VF("/VOUT"))	
1	ITI_LABS:LAB_08_tb:1	ymax(dB20(VF("/VOUT")))	5.757
1	ITI_LABS:LAB_08_tb:1	ymax(mag(VF("/VOUT")))	1.94
1	ITI_LABS:LAB_08_tb:1	bandwidth(VF("/VOUT") 3 "low")	2.573M
1	ITI_LABS:LAB_08_tb:1	gainBwProd(VF("/VOUT"))	5.005M
1	ITI_LABS:LAB_08_tb:1	unityGainFreq(VF("/VOUT"))	4.314M
Parameters: CIN=12p			
2	ITI_LABS:LAB_08_tb:1	dB20(VF("/VOUT"))	
2	ITI_LABS:LAB_08_tb:1	ymax(dB20(VF("/VOUT")))	11.52
2	ITI_LABS:LAB_08_tb:1	ymax(mag(VF("/VOUT")))	3.768
2	ITI_LABS:LAB_08_tb:1	bandwidth(VF("/VOUT") 3 "low")	1.327M
2	ITI_LABS:LAB_08_tb:1	gainBwProd(VF("/VOUT"))	5.011M
2	ITI_LABS:LAB_08_tb:1	unityGainFreq(VF("/VOUT"))	4.848M

Figure 5: All Simulation Results for AC closed loop

Table 1: Simulation Results

Parameters	Cin = 4pF	Cin = 12pF
DC Gain in dB	5.757	11.52
DC Gain	1.94	3.768
BW	2.573 MHz	1.327 MHz
GBW	5 MHz	5 MHz
UGF	4.314 MHz	4.848 MHz

1.1.2 *Hand Analysis:*

From the Lab 7 simulation results, the behavioral OTA parameters are modeled as follows:

- Open-Loop DC Gain (A_{OL}) = 36.13 dB (65 V/V)
- Open-Loop Bandwidth (BW_{OL}) = 76.1 kHz
- Open-Loop Gain-Bandwidth Product (GBW_{OL}) = 4.9 MHz

$$Ideal\ \text{closed loop}\ Gain = 1 + \frac{C_{IN}}{C_F}$$

The feedback factor :

$$\beta = \frac{C_F}{C_F + C_{IN}}$$

So

$$A_{CL} = \frac{A_{OL}}{1 + \beta A_{OL}}$$

$$BW_{CL} = BW_{OL} \times (1 + \beta A_{OL})$$

For C_{IN} = 4pF:

- $\beta = \frac{4\text{pF}}{4\text{pF}+4\text{pF}} = 0.5$

- Ideal Gain $= 1 + \frac{4\text{pF}}{4\text{pF}} = 2$ V/V (6.02 dB)
- $A_{CL} = \frac{65}{1+(0.5\times 65)} = 1.94$ V/V (5.75 dB)
- $BW_{CL} = 76.1$ kHz $\times (1 + 32.5) = 2.55$ MHz

For $C_{IN} = 12\text{pF}$:

- $\beta = \frac{4\text{pF}}{4\text{pF}+12\text{pF}} = 0.25$
- Ideal Gain $= 1 + \frac{12\text{pF}}{4\text{pF}} = 4$ V/V (12.04 dB)
- $A_{CL} = \frac{65}{1+(0.25\times 65)} = 3.768$ V/V (11.51 dB)
- $BW_{CL} = 76.1$ kHz $\times (1 + 16.25) = 1.31$ MHz

Table 2: Closed-Loop AC Parameters – Hand Calculation vs. Simulation

Parameter	Cin = 4PF		Cin = 12pF	
	Simulation	Analytical	Simulation	Analytical
DC Gain in dB	5.757 dB	5.75 dB	11.52 dB	11.51 dB
DC Gain	1.94	1.94	3.768	3.764
BW	2.573 MHz	2.51 MHz	1.327 MHz	1.3 MHz
GBW	5 MHz	4.95 MHz	5 MHz	4.94 MHz

Comment on the difference between the results for the two values of CIN.

Increasing Cin decreases the feedback factor to the half which in approx. it doubles the closed loop gain while halving the BW and the GBW is unchanged.

1.2 Loop gain vs frequency (STB Analysis):

1.2.1 Simulation Results:

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	ITI_LABS:LAB_08_tb:1	unityGainFreq(mag(getData("loopGain" ?result "stb")))	2.5M
1	ITI_LABS:LAB_08_tb:1	ymax(mag(getData("loopGain" ?result "stb")))	32.5
1	ITI_LABS:LAB_08_tb:1	ymax(dB20(getData("loopGain" ?result "stb")))	30.24
Parameters: CIN=12p			
2	ITI_LABS:LAB_08_tb:1	unityGainFreq(mag(getData("loopGain" ?result "stb")))	1.249M
2	ITI_LABS:LAB_08_tb:1	ymax(mag(getData("loopGain" ?result "stb")))	16.25
2	ITI_LABS:LAB_08_tb:1	ymax(dB20(getData("loopGain" ?result "stb")))	24.22

Figure 6: STB Analysis Results

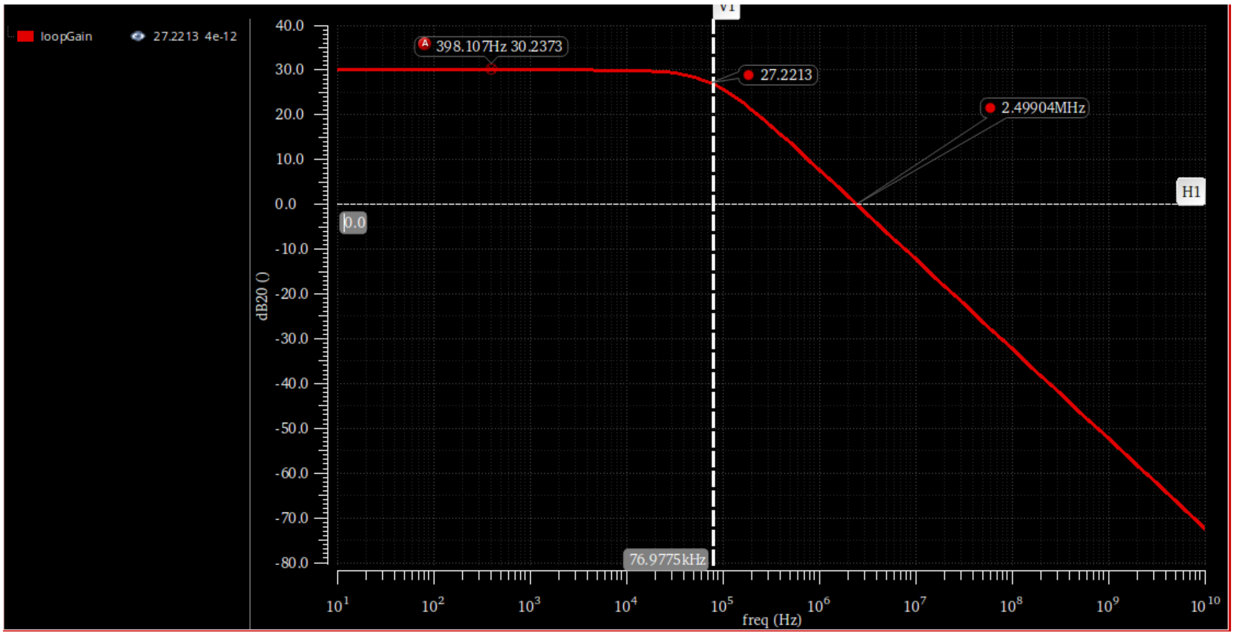

Figure 7: Loop gain magnitude vs. frequency for C_{IN} = 4pF

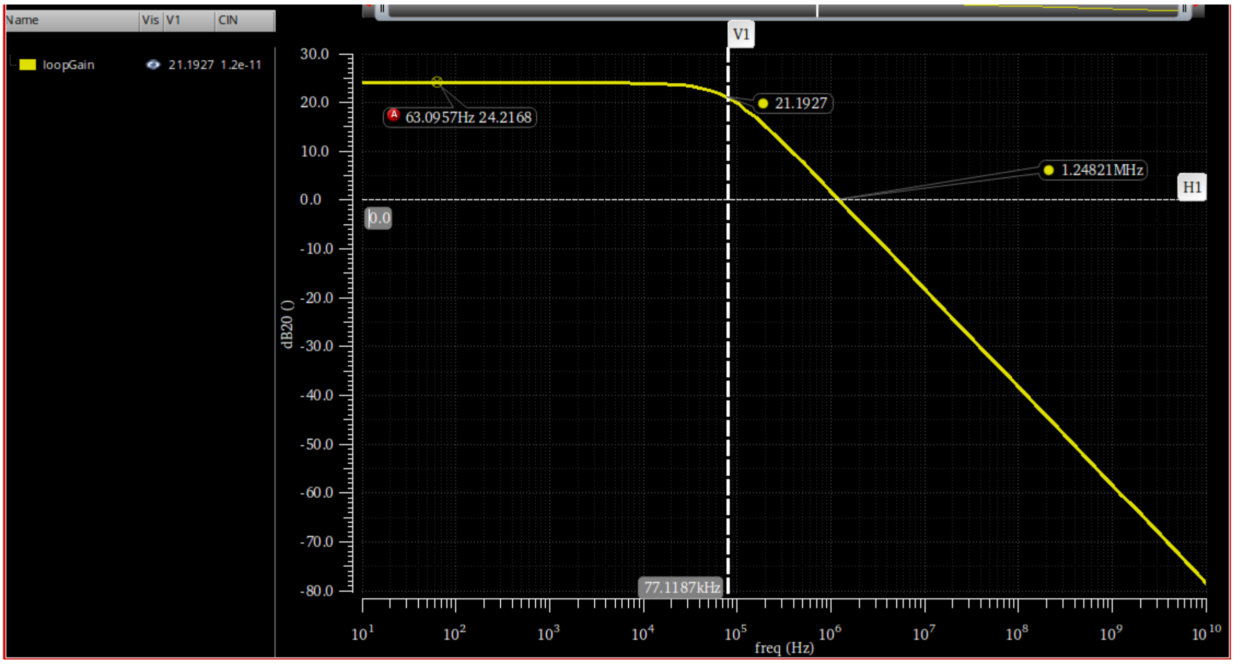

Figure 8: Loop gain magnitude vs. frequency for C_{IN} = 12pF

1.2.2 Hand Analysis:

From the Lab 7 open-loop parameters, we have:

- Open-Loop DC Gain (A_{OL}) = 65 V/V
- Open-Loop Bandwidth (BW_{OL}) = 76.1 kHz
- Open-Loop Gain-Bandwidth Product (GBW_{OL}) = 4.9 MHz

$$LG = \beta \times A_{OL}$$

The dominant pole is the same as open loop pole:

$$Pole_{dominant} = BW_{OL} = 76.1\ KHz$$

The UGF is scaled with feedback factor

$$UGF_{LG} = \beta \times GBW_{OL}$$

For C_{IN} = 4pF:

- $\beta = 0.5$
- $LG = 0.5 \times 65 = 32.5$ V/V (30.24 dB)
- $Pole_{dominant} = 76.1$ kHz
- $UGF_{LG} = 0.5 \times 4.9$ MHz $= 2.45$ MHz

For C_{IN} = 12pF:

- $\beta = 0.25$
- $LG = 0.25 \times 65 = 16.25$ V/V (24.22 dB)
- $Pole_{dominant} = 76.1$ kHz
- $UGF_{LG} = 0.25 \times 4.9$ MHz $= 1.225$ MHz

Table 3: Loop Gain Parameters – Hand Calculation vs. Simulation

Parameter	Cin = 4PF		Cin = 12pF	
	Simulation	**Analytical**	**Simulation**	**Analytical**
DC Loop Gain in dB	30.24 dB	30.24 dB	24.22 dB	24.22 dB
DC Loop Gain	32.5	32.51	16.25	16.255
BW	77 MHz	76.1 MHz	76.9 MHz	76.1 MHz
UGF	2.5 MHz	2.45 MHz	1.25 MHz	1.225 MHz

Comment on the differences between the results for the two values of C_{IN}.

Increasing C_{IN} from 4pF to 12pF reduces the feedback factor β by half, halving the feedback factor causes the DC loop gain to drop by a factor of 2 (a 6 dB decrease) and also halves the unity gain frequency (UGF) of the loop. However, because the feedback network is an ideal capacitive voltage divider, it does not alter the frequency response characteristics of the OTA itself, meaning the dominant pole remains constant at approximately 76.1 kHz for both cases.

1.3 Gain Desensitization:

1.3.1 Simulation Results:

As I don't have Av as a variable in the behavioral circuit I only have GM and Rout so I will sweep over one of them

$$A_{OL} = GM \times ROUT$$

To sweep over GM, Rout = 414 k

Start: $\mathrm{GM} = \frac{Av}{ROUT} = \frac{50}{414\ k} = 120\ \mu A/V\ (10^{-4}\ S)$

End : $\mathrm{GM} = \frac{Av}{ROUT} = \frac{50000}{414\ k} = 120\ mA/V\ (10^{-4}\ S)$

To sweep over Rout, GM = 157 u

Start: $\mathrm{ROUT} = \frac{Av}{GM} = \frac{50}{157\ u} = 320\ K\Omega$

End: $\mathrm{ROUT} = \frac{Av}{GM} = \frac{50000}{157\ u} = 320\ M\Omega$

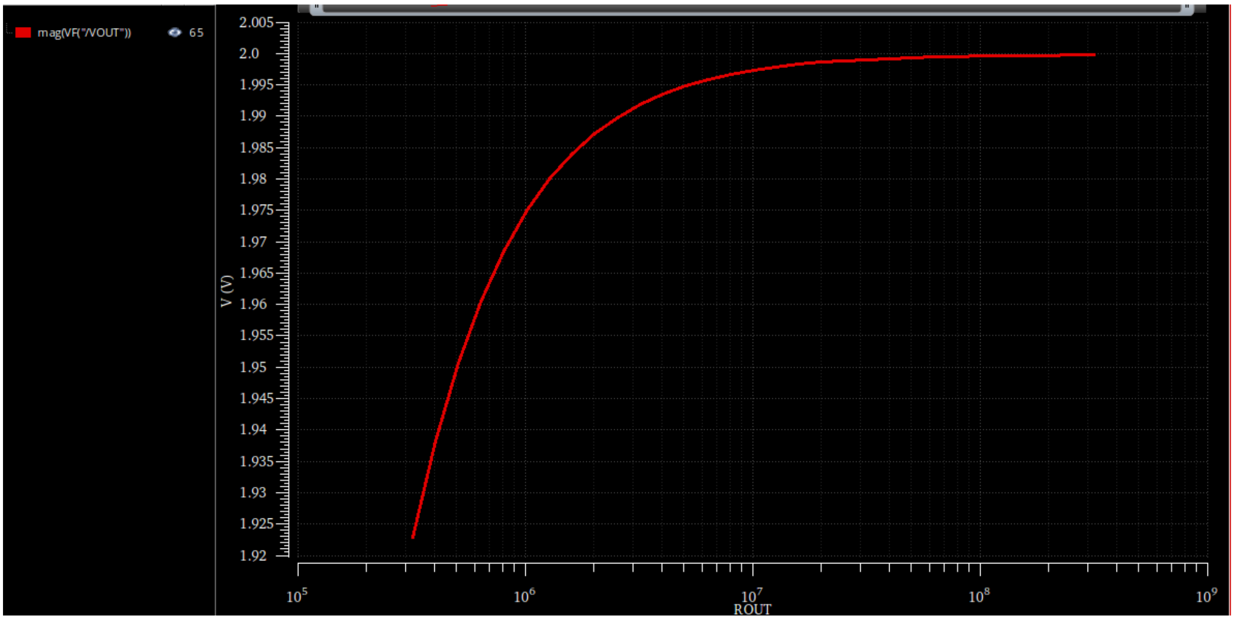

Figure 9: Closed-loop DC gain magnitude vs. ROUT sweep.

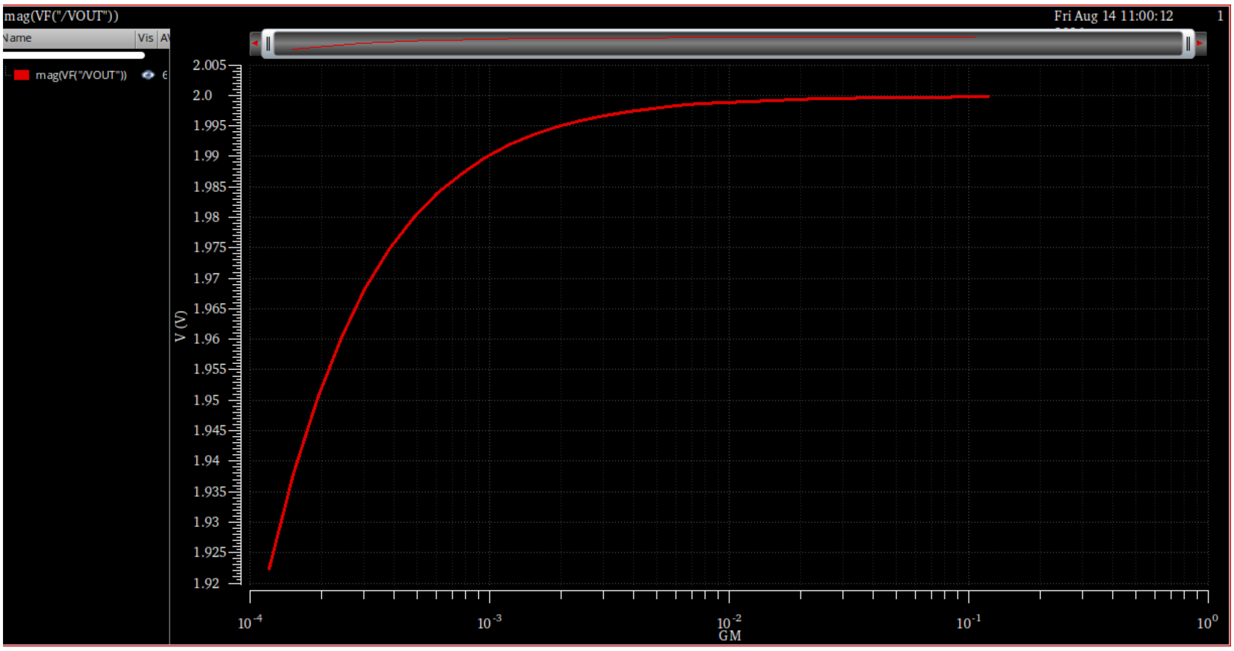

Figure 10: Closed-loop DC gain magnitude vs. GM sweep

1.3.2 Hand Analysis:

$$A_{CL} = \frac{A_{OL}}{1 + \beta A_{OL}}$$

$A_{cl} = 50/26 = 1.923$

$A_{cl} = 50000/25001 = 1.99996$

$$\% \Delta A_{cl} = \frac{1.99996 - 1.923}{1.923} \times 100\% \approx 4.0\%$$

Just a change 4 percent while the open loop changed 1000 times which is 60dB, the closed-loop gain sensitivity to open-loop gain is suppressed by roughly the same factor as the loop gain itself this is the core of desensitization: $\frac{dA_{cl}/A_{cl}}{dA_v/A_v} = \frac{1}{1+\beta A_v}$, which shrinks toward zero as loop gain grows.

2 PART 2: Feedback with Real 5T OTA

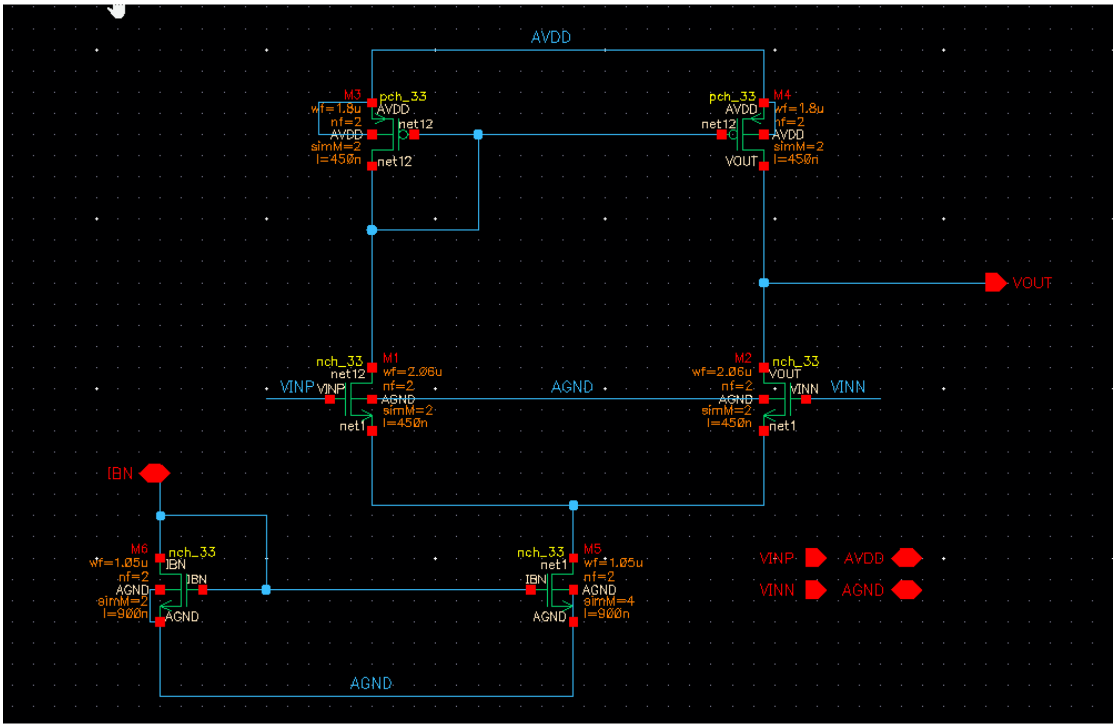

Figure 11: Real 5T OTA schematic

2.1 Closed-Loop Gain vs. Frequency

2.1.1 Simulation Results:

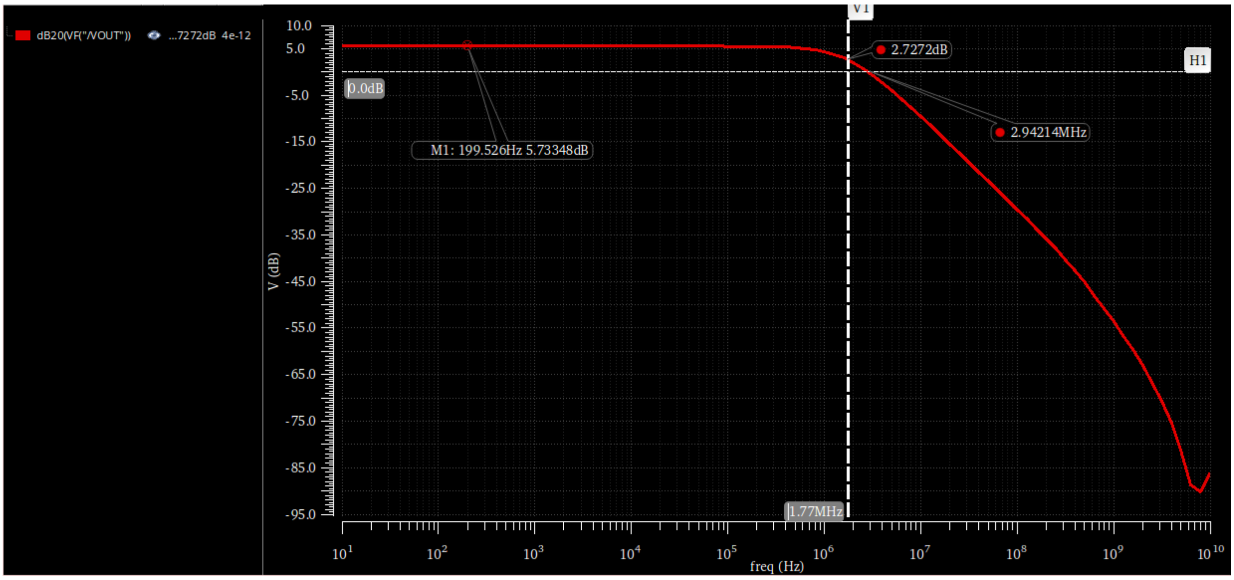

Figure 12: VOUT in dB with Cin=4pF

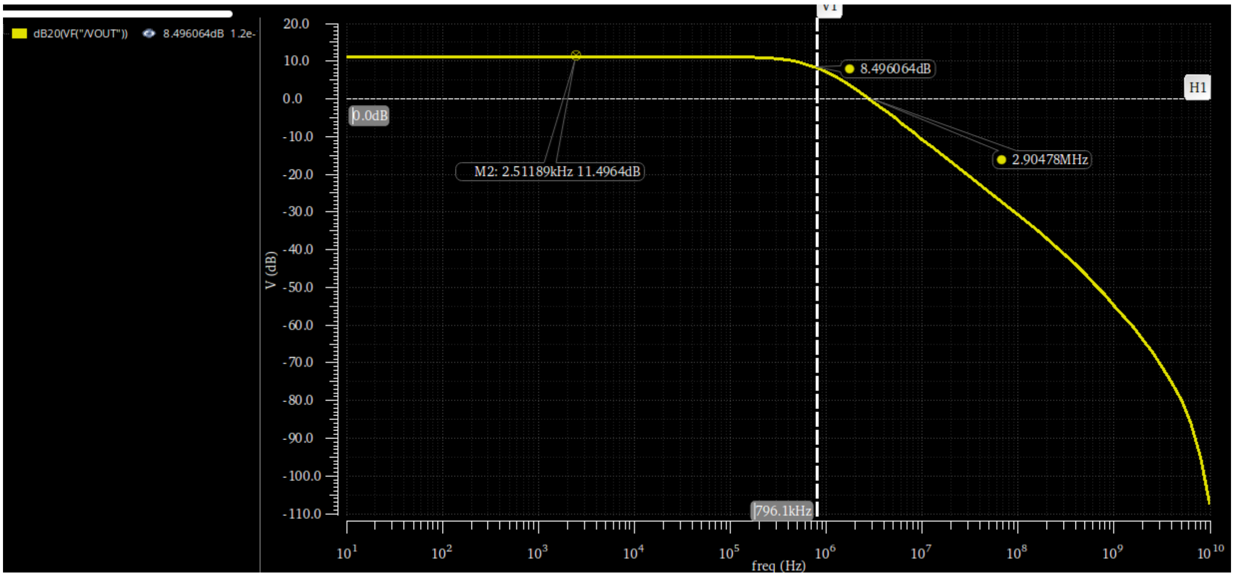

Figure 13: VOUT in dB with Cin=12pF

Parameters: CIN=4p			
1	ITI_LABS:LAB_08_tb_2:1	dB20(VF("/VOUT"))	
1	ITI_LABS:LAB_08_tb_2:1	ymax(dB20(VF("/VOUT")))	5.733
1	ITI_LABS:LAB_08_tb_2:1	ymax(mag(VF("/VOUT")))	1.935
1	ITI_LABS:LAB_08_tb_2:1	bandwidth(VF("/VOUT") 3 "low")	1.773M
1	ITI_LABS:LAB_08_tb_2:1	gainBwProd(VF("/VOUT"))	3.439M
1	ITI_LABS:LAB_08_tb_2:1	unityGainFreq(VF("/VOUT"))	2.954M
Parameters: CIN=12p			
2	ITI_LABS:LAB_08_tb_2:1	dB20(VF("/VOUT"))	
2	ITI_LABS:LAB_08_tb_2:1	ymax(dB20(VF("/VOUT")))	11.5
2	ITI_LABS:LAB_08_tb_2:1	ymax(mag(VF("/VOUT")))	3.757
2	ITI_LABS:LAB_08_tb_2:1	bandwidth(VF("/VOUT") 3 "low")	796.1k
2	ITI_LABS:LAB_08_tb_2:1	gainBwProd(VF("/VOUT"))	2.999M
2	ITI_LABS:LAB_08_tb_2:1	unityGainFreq(VF("/VOUT"))	2.921M

Figure 14: All Simulation Results for AC closed loop

Table 4: Closed-Loop AC Parameters – Behavioral vs Real OTA

Parameter	Cin = 4PF		Cin = 12pF	
	Behavioral OTA	**Real OTA**	**Behavioral OTA**	**Real OTA**
DC Gain in dB	5.757 dB	5.733 dB	11.52 dB	11.5 dB
DC Gain	1.94	1.94	3.768	3.757
BW	2.573 MHz	1.773 MHz	1.327 MHz	796.1 kHz
UGF	4.314 MHz	2.954 MHz	4.85 MHz	2.921 MHz
GBW	5 MHz	3.439 MHz	5 MHz	3 MHz

Comment on the differences between Behavioral and Real OTA results.

The simulated closed-loop DC gains for the real 5T OTA matches the behavioral model. However, the bandwidth and Gain-Bandwidth Product (GBW) are smaller in the real OTA and this was expected. This occurs because the behavioral model utilized an ideal buffer at its output, which completely prevented the capacitive feedback network from loading the OTA. Also in the real 5T OTA (C_F in series with C_{IN}) directly loads the output node, adding to the existing load capacitance (C_L) and the parasitic & internal caps of MOSFETS. This increased total capacitive loading lowers the dominant pole frequency, reducing both the closed-loop bandwidth and the overall GBW.

2.2 Loop gain vs frequency (STB Analysis):

2.2.1 Simulation Results:

Parameters: CIN=4p			
1	ITI_LABS:LAB_08_tb_2:1	unityGainFreq(mag(getData("loopGain" ?result "stb")))	1.735M
1	ITI_LABS:LAB_08_tb_2:1	ymax(mag(getData("loopGain" ?result "stb")))	32.16
1	ITI_LABS:LAB_08_tb_2:1	ymax(dB20(getData("loopGain" ?result "stb")))	30.15
1	ITI_LABS:LAB_08_tb_2:1	dB20(getData("loopGain" ?result "stb"))	
Parameters: CIN=12p			
2	ITI_LABS:LAB_08_tb_2:1	unityGainFreq(mag(getData("loopGain" ?result "stb")))	756.4k
2	ITI_LABS:LAB_08_tb_2:1	ymax(mag(getData("loopGain" ?result "stb")))	16.19
2	ITI_LABS:LAB_08_tb_2:1	ymax(dB20(getData("loopGain" ?result "stb")))	24.18
2	ITI_LABS:LAB_08_tb_2:1	dB20(getData("loopGain" ?result "stb"))	

Figure 15: STB Analysis Results

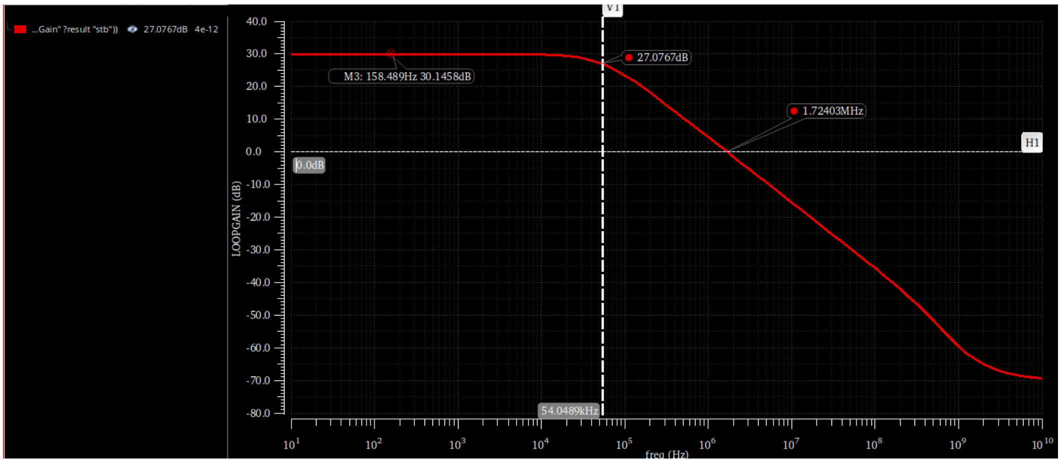

Figure 16: Loop gain magnitude vs. frequency for C_{IN} = 4pF

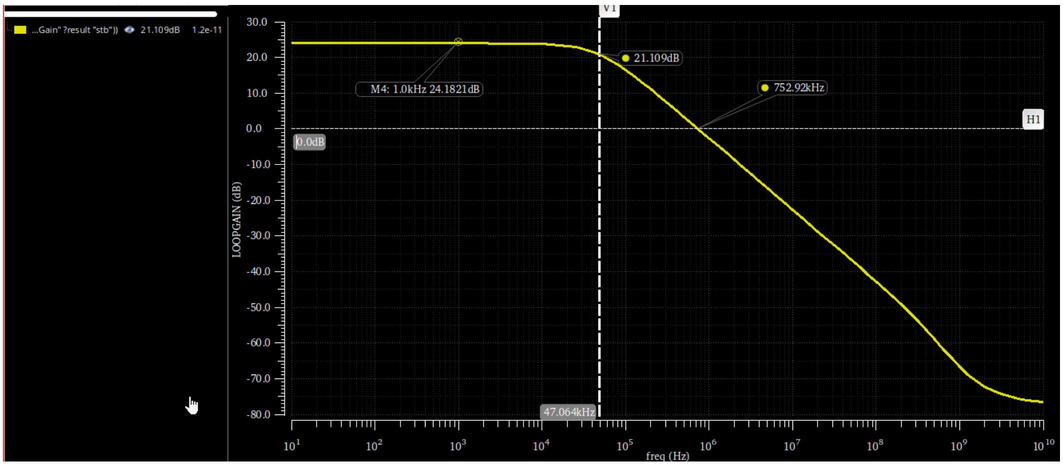

Figure 17: Loop gain magnitude vs. frequency for C_{IN} = 12pF

Table 5: Loop Gain Parameters – Behavioral vs. Real OTA

Parameter	Cin = 4PF		Cin = 12pF	
	Behavioral	**Real**	**Behavioral**	**Real**
DC Loop Gain in dB	30.24 dB	30.15 dB	24.22 dB	24.18 dB
DC Loop Gain	32.5	32.16	16.25	16.19
BW	77 MHz	54 MHz	76.9 MHz	47 MHz
UGF	2.5 MHz	1.735 MHz	1.25 MHz	756.4 kHz

Comment on the differences between Behavioral and Real OTA results:

The unity gain frequency (UGF) and the dominant pole of the loop gain are smaller in the real 5T OTA compared to the behavioral model. This is due to the same loading effect observed in the closed-loop AC response. The behavioral model uses an ideal buffer that isolates the feedback network (C_F and C_{IN}) from the OTA's output. In the real transistor-level design, the feedback network acts as an additional capacitive load connected to the output node. Because the dominant pole is inversely proportional to the total output capacitance ($f_p \propto 1/C_{total}$), this extra loading shifts the dominant pole to a lower frequency (dropping from 76.1 kHz down to the 47-54 kHz range). Since the loop's UGF is approximately the product of the feedback factor (β) and the open-loop Gain-Bandwidth Product (GBW), the reduced GBW from the heavier capacitive load directly results in a proportionally smaller loop gain UGF.

2.3 Gain Desensitization:

2.3.1 Simulation Results:

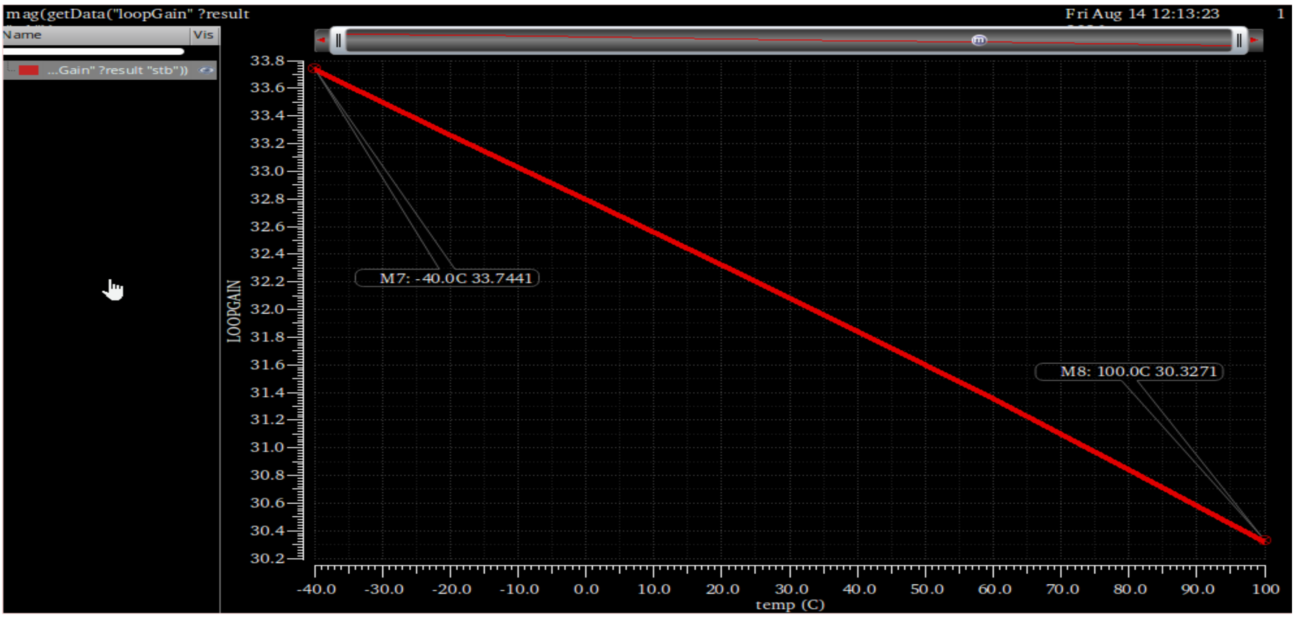

Figure 18: DC loop gain magnitude (linear) vs. temperature sweep

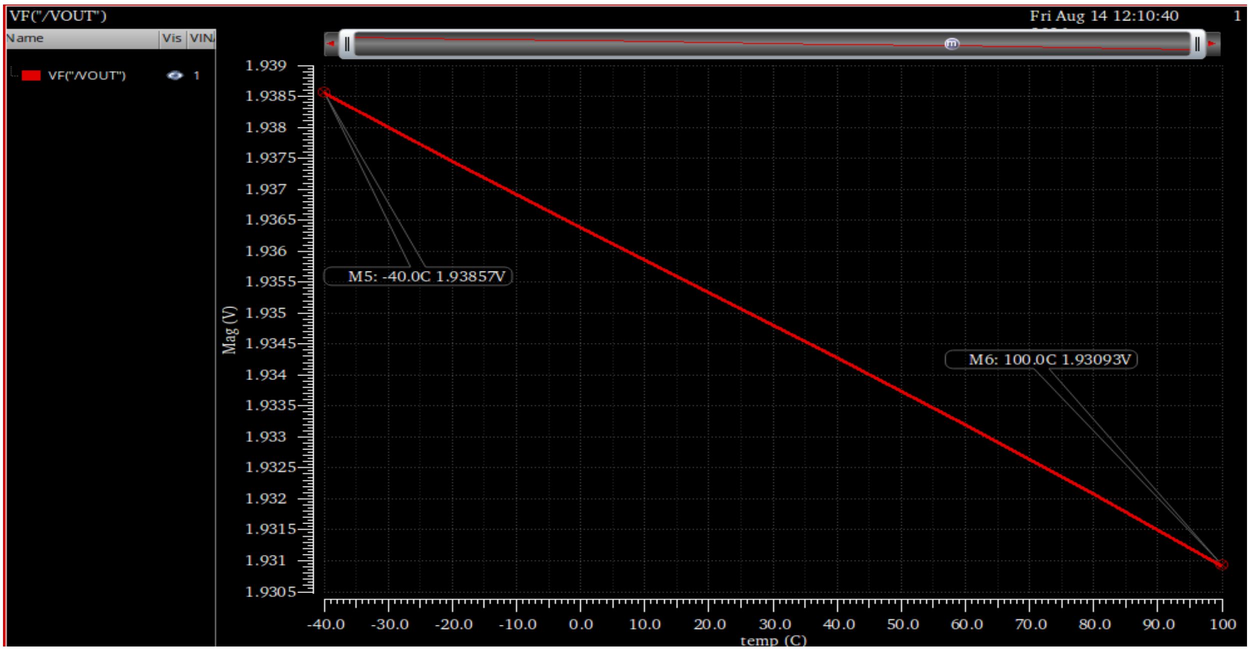

Figure 19: Closed-loop DC gain magnitude (linear) vs. temperature sweep.

From the STB analysis (Loop Gain):

- Maximum Loop Gain (at -40°C): 33.7441 V/V
- Minimum Loop Gain (at 100°C): 30.3271 V/V
- Percent Change: $\frac{33.7441-30.3271}{30.3271} \times 100 = 11.27\%$

From the AC analysis (Closed-Loop Gain):

- Maximum Closed-Loop Gain (at -40°C): 1.93857 V/V
- Minimum Closed-Loop Gain (at 100°C): 1.93093 V/V
- Percent Change: $\frac{1.93857-1.93093}{1.93093} \times 100 = 0.40\%$

Comment on Gain Desensitization with Temperature:

The simulation results clearly show the gain desensitization of negative feedback under physical variations. Temperature impact transistor characteristics (mobility and threshold voltage), causing the open-loop gain and the loop gain of the real 5T OTA to vary by 11.27% across the swept range. However, because the system is placed in closed-loop negative feedback, this is suppressed, resulting in a 0.40% change in the actual closed-loop gain. This confirms that the closed-loop performance is highly stable and defined by the passive feedback components rather than the active amplifier variations.

2.4 Transient analysis:

2.4.1 *Simulation Results at $F_{IN} = 1$ kHz:*

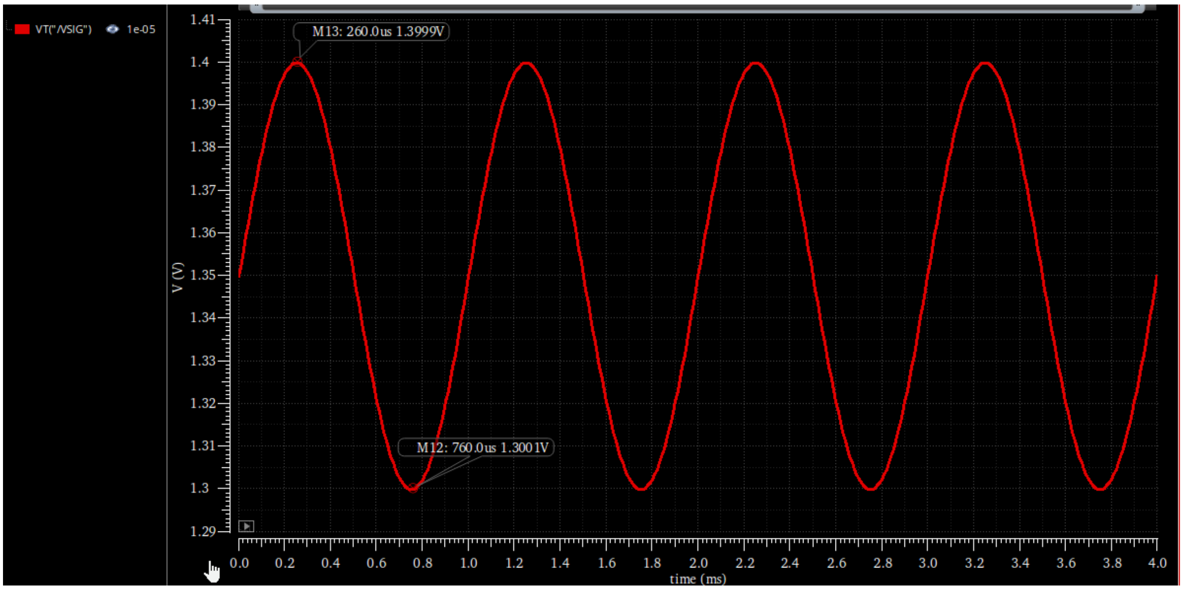

Figure 20: Transient input signal (V_{IN}) at 1 kHz.

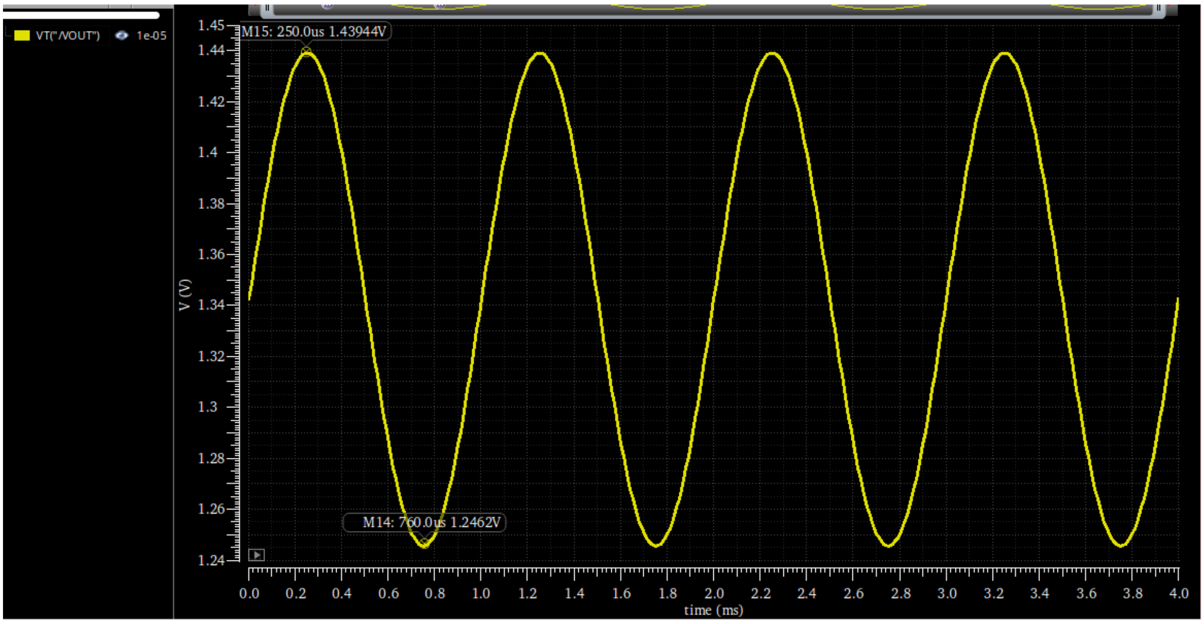

Figure 21: Transient output signal (V_{OUT}) at 1 kHz.

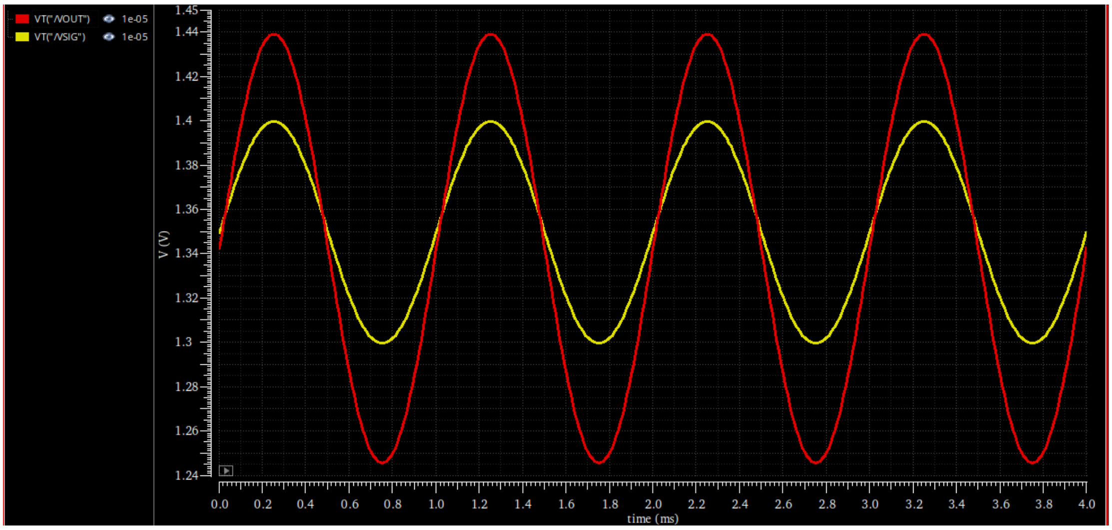

Figure 22: Transient input and output signals overlaid at 1 *kHz.*

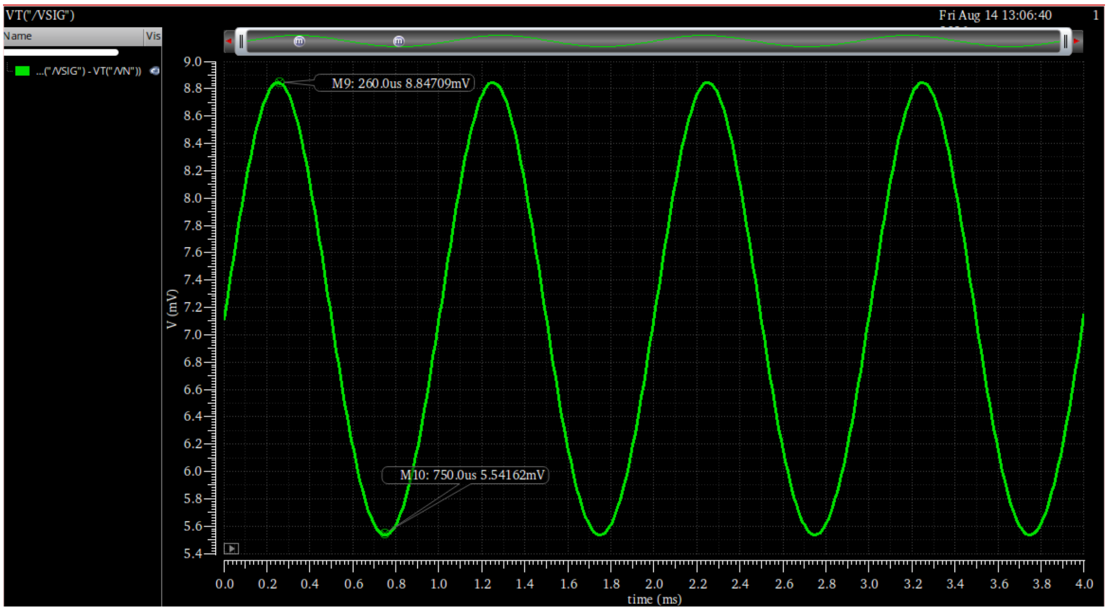

Figure 23: Differential input signal ($V_P - V_N$) *at* 1 *kHz.*

Input V_{IN} $(V_{pp}) =$ 100.0 mV

Output V_{OUT} $(V_{pp}) =$ 193.24 mV

Differential Input $V_P - V_N$ $(V_{pp}) \approx 3.31$ mV

2.4.2 Simulation Results at $F_{IN} = BW_{CL}$ (1.773 MHz):

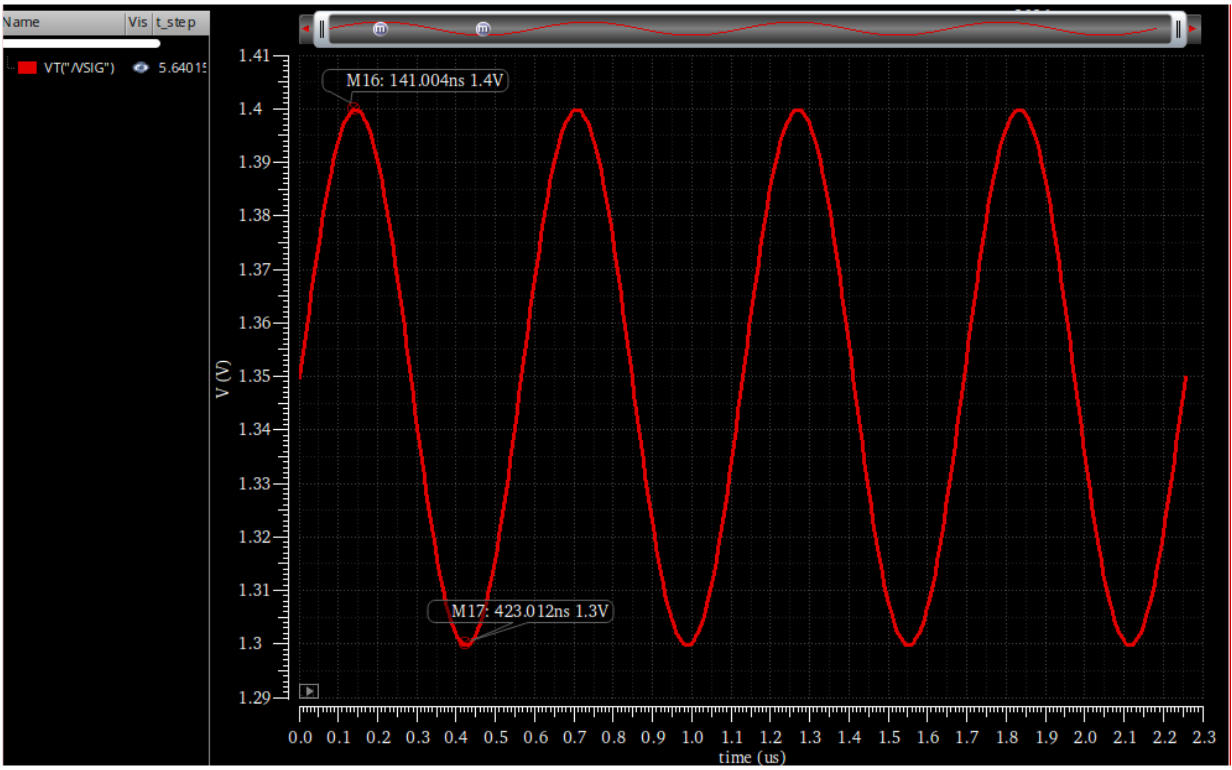

Figure 24: Transient input signal (V_{IN}) at 1.773 MHz

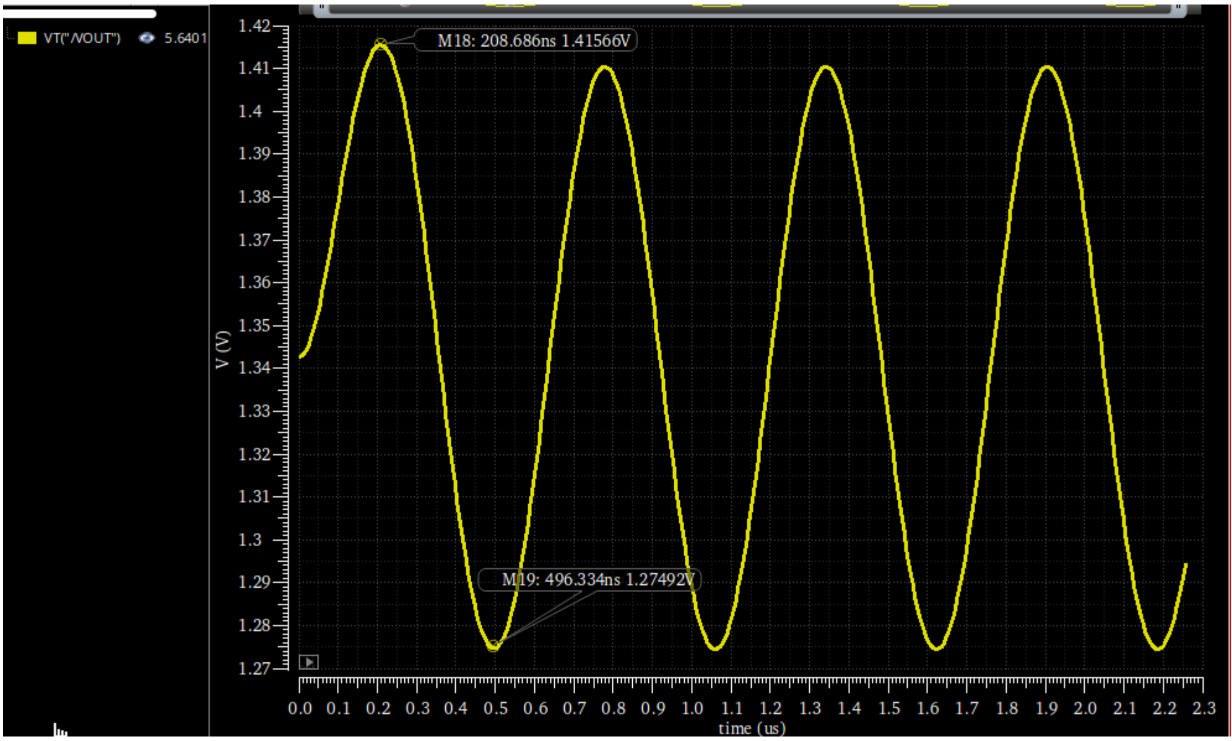

Figure 25: Transient output signal (V_{OUT}) at 1.773 MHz.

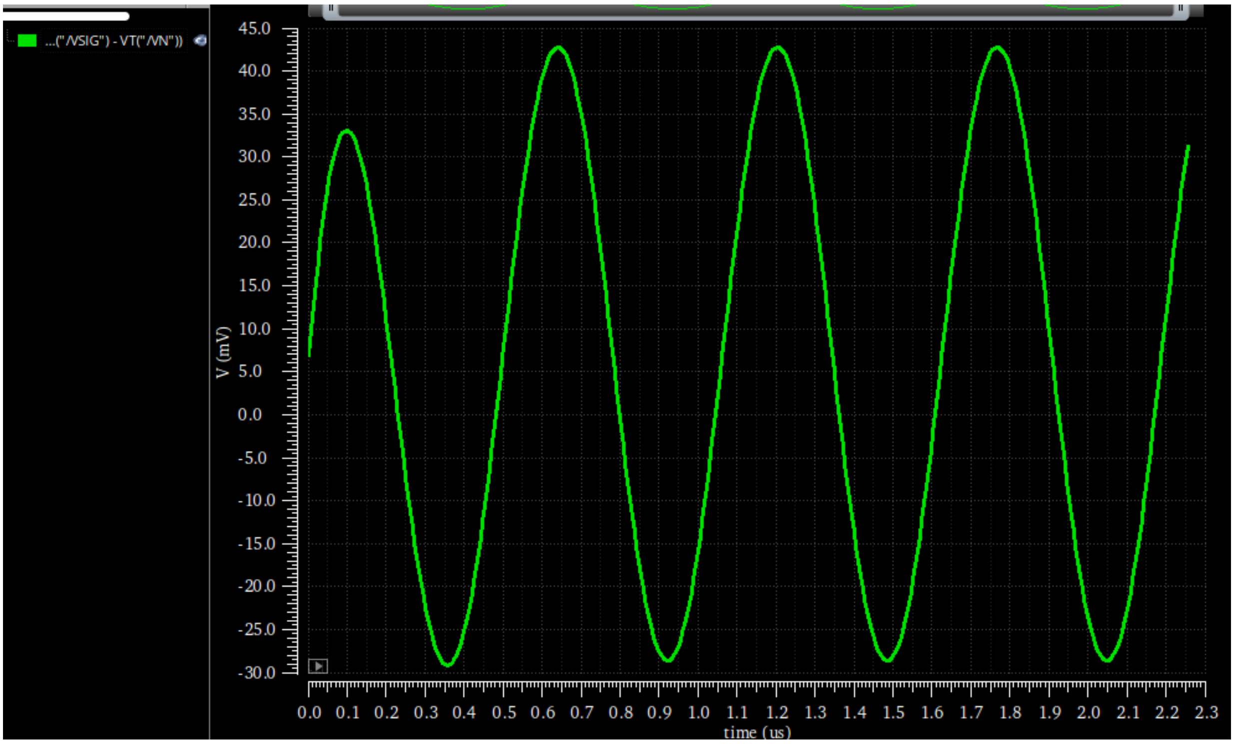

Figure 26: Differential input signal ($V_P - V_N$) at 1.773 MHz.

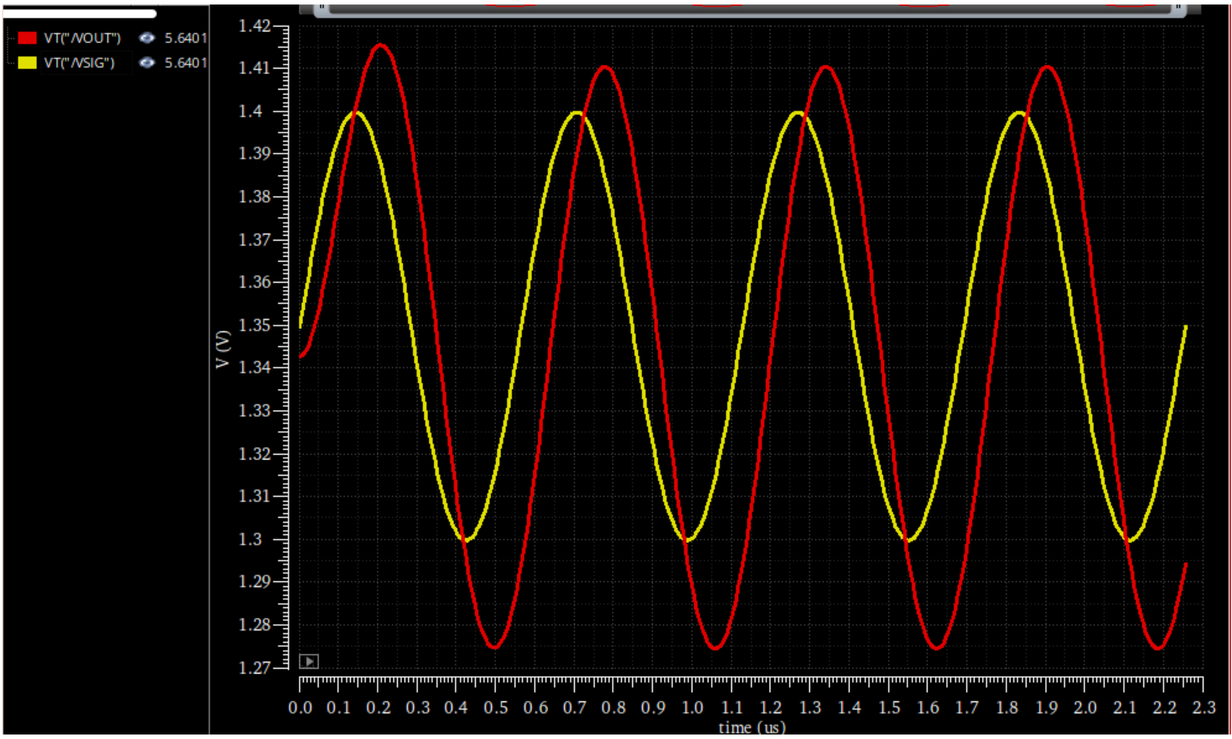

Figure 27: Transient input and output signals overlaid at 1.773 MHz.

Input V_{IN} $(V_{pp}) =$ 100.0 mV

Output V_{OUT} $(V_{pp}) =$ 140.74 mV

Differential Input $V_P - V_N$ $(V_{pp}) \approx 72.0$ mV

Relation	FIN = 1 kHz	FIN = 1.773 MHz
Closed-Loop Gain ($V_{OUT,pp}/V_{IN,pp}$)	1.93	1.41
Open-Loop Gain ($V_{OUT,pp}/V_{vid,pp}$)	58.4	1.95

Comment on Transient Relations:

The relation between the output signal and the differential input signal $(V_P - V_N)$ represents the open-loop gain of the OTA:

- **At 1 kHz:** The OTA operates well within its bandwidth, exhibiting its full open-loop DC gain (calculated here as 58.4 V/V). Because the open-loop gain is high, the negative feedback is very strong, forcing the differential input voltage $(V_P - V_N)$ to remain extremely small ($\approx$ 3.31 mV). The resulting closed-loop gain is 1.93 V/V, which tracks the ideal theoretical gain of 2 V/V.
- **At 1.773 MHz (BW_{CL}):** At this much higher frequency, the open-loop gain of the OTA rolls off significantly due to the dominant pole. With the open-loop gain dropping to approximately 1.95 V/V, the negative feedback loop loses its effectiveness. As a result, the virtual short between the input terminals breaks down, causing the differential input voltage $(V_P - V_N)$ to swing massively to $\approx$ 72.0 mV. The closed-loop gain drops to 1.41 V/V (which is $1.93/\sqrt{2}$). At this high frequency, the open-loop and closed-loop asymptotes coincide, leading to the open-loop and closed-loop gains becoming almost identical.